THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 8:12:15	Make B equal: B $= \text{LCM}(3,4) = 12$; A $= 8$, C $= 15$ gives $8 : 12 : 15$
2	**(a)** 15	$18 \times 10 = 12 \times x$
3	**(b)** 9	$135 \div 15$
4	**(d)** 84	$14/60 \times 360$
5	**(a)** 900	One part $= ₹300$
6	**(d)** 560	Speed $= 70$ km/h
7	**(d)** 5:8	Compare as fractions: $\frac{5}{8} \approx 0.625$, $\frac{7}{12} \approx 0.583$, $\frac{3}{5} \approx 0.600$, $\frac{4}{7} \approx 0.571$
8	**(c)** 15:20:24	Make B equal: B $= \text{LCM}(4,5) = 20$; A $= 15$, C $= 24$ gives $15 : 20 : 24$
9	**(c)** 12	$20 \times 9 = 15 \times x$
10	**(c)** 7	$175 \div 25$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{5}{9} = 0.56$ and $\frac{3}{5} = 0.6$ (1); so 3:5 is greater.	1+1
12	$xy = 6 \times 20 = 120$ (1); $y = 120 \div 8 = 15$ (1).	1+1
13	$k = y/x = 35/7 = 5$ (1); $y = 5 \times 12 = 60$ (1).	1+1
14	$\frac{8}{15} = 0.53$ and $\frac{7}{13} = 0.54$ (1); so 7:13 is greater.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Length $= 14$ km (1); breadth $= 8$ km (1); perimeter $= 2(14 + 8) = 44$ km (1).	1+1+1
16	$1\% = 3.6°$ (1); food $108°$, rent $90°$ (1); education $72°$, others $90°$ (1).	1+1+1
17	Speed × time is constant: $30 \times 24 = 720$ (1); new speed $= 720 \div 18$ (1) $= 40$ km/h (1).	1+1+1
	OR: Length $= 18$ km (1); breadth $= 15$ km (1); perimeter $= 2(18 + 15) = 66$ km (1).	1+1+1
18	$1\% = 3.6°$ (1); food $126°$, rent $72°$ (1); education $54°$, others $108°$ (1).	1+1+1
	OR: Speed × time is constant: $36 \times 25 = 900$ (1); new speed $= 900 \div 20$ (1) $= 45$ km/h (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Make B equal: A:B:C $= 8 : 12 : 15$ (2); total parts $= 35$, one part $= ₹150$ (1); A = ₹ 1200, B = ₹ 1800, C = ₹ 2250 (2).	2+1+2
	OR: Angle per mark $= 360° \div 60 = 6°$ (1); angles: Maths 108°, Science 90°, English 54°, Hindi 72°, SST 36° (3); Maths percentage $= 30\%$ (1).	1+3+1
20	Make B equal: A:B:C $= 3 : 5 : 7$ (2); total parts $= 15$, one part $= ₹100$ (1); A = ₹ 300, B = ₹ 500, C = ₹ 700 (2).	2+1+2
	OR: Angle per mark $= 360° \div 60 = 6°$ (1); angles: Maths 72°, Science 144°, English 60°, Hindi 48°, SST 36° (3); Maths percentage $= 20\%$ (1).	1+3+1

Mistakes to avoid:

- Treating inverse proportion as direct: more workers means **fewer** days, so use $x_1 y_1 = x_2 y_2$.
- Not converting the map scale to the same units (cm to km) before computing.
- Dividing instead of multiplying (or the other way) when converting map distance to actual distance.
- Computing the pie-chart angle as $\frac{\text{value}}{\text{total}} \times 100$ (percentage) instead of $\times 360°$.
- Using direct proportion for quantities that are not proportional (e.g. age and height).
- Not checking that all sector angles add up to $360°$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 144	$18/45 \times 360$
2	**(a)** 1200	One part $=$ ₹300
3	**(a)** 450	Speed $=$ 45 km/h
4	**(d)** 4:9	Compare as fractions: $\frac{3}{7} \approx 0.429$, $\frac{4}{9} \approx 0.444$, $\frac{5}{12} \approx 0.417$, $\frac{2}{5} \approx 0.400$
5	**(d)** 3:6:8	Make B equal: B $= \text{LCM}(2,3) = 6$; A $= 3$, C $= 8$ gives $3 : 6 : 8$
6	**(c)** 10	$16 \times 15 = 24 \times x$
7	**(a)** 8	$144 \div 18$
8	**(c)** 105	$35/120 \times 360$
9	**(a)** 1000	One part $=$ ₹200
10	**(a)** 396	Speed $=$ 44 km/h

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$xy = 9 \times 16 = 144$ (1); $y = 144 \div 12 = 12$ (1).	1+1
12	$k = y/x = 48/6 = 8$ (1); $y = 8 \times 9 = 72$ (1).	1+1
13	$\frac{11}{12} = 0.92$ and $\frac{9}{10} = 0.9$ (1); so 11:12 is greater.	1+1
14	$xy = 15 \times 8 = 120$ (1); $y = 120 \div 6 = 20$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Length $=$ 45 km (1); breadth $=$ 20 km (1); perimeter $= 2(45 + 20) = 130$ km (1).	1+1+1
16	$1\% = 3.6°$ (1); food $144°$, rent $54°$ (1); education $36°$, others $126°$ (1).	1+1+1
17	Speed × time is constant: $40 \times 18 = 720$ (1); new speed $= 720 \div 15$ (1) $= 48$ km/h (1).	1+1+1
	OR: Length $=$ 32 km (1); breadth $=$ 12 km (1); perimeter $= 2(32 + 12) = 88$ km (1).	1+1+1
18	$1\% = 3.6°$ (1); food $90°$, rent $108°$ (1); education $90°$, others $72°$ (1).	1+1+1
	OR: Speed × time is constant: $45 \times 16 = 720$ (1); new speed $= 720 \div 12$ (1) $= 60$ km/h (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Make B equal: A:B:C $= 3 : 6 : 8$ (2); total parts $= 17$, one part $=$ ₹120 (1); A = ₹ 360, B = ₹ 720, C = ₹ 960 (2).	2+1+2
	OR: Angle per mark $= 360° \div 40 = 9°$ (1); angles: Wheat 126°, Rice 90°, Maize 72°, Bajra 54°, Pulses 18° (3); Wheat percentage $= 35\%$ (1).	1+3+1
20	Make B equal: A:B:C $= 2 : 5 : 6$ (2); total parts $= 13$, one part $=$ ₹90 (1); A = ₹ 180, B = ₹ 450, C = ₹ 540 (2).	2+1+2
	OR: Angle per mark $= 360° \div 30 = 12°$ (1); angles: Tea 72°, Coffee 60°, Milk 120°, Juice 60°, Water 48° (3); Tea percentage $= 20\%$ (1).	1+3+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 9:11	Compare as fractions: $\frac{9}{11} \approx 0.818$, $\frac{13}{16} \approx 0.812$, $\frac{4}{5} \approx 0.800$, $\frac{7}{9} \approx 0.778$
2	**(a)** 5:6:9	Make B equal: B $= \text{LCM}(6,2) = 6$; A $= 5$, C $= 9$ gives $5:6:9$
3	**(a)** 8	$28 \times 6 = 21 \times x$
4	**(d)** 8	$360 \div 45$
5	**(c)** 55	$11/72 \times 360$
6	**(d)** 900	One part $=$ ₹300
7	**(c)** 455	Speed $= 65$ km/h
8	**(a)** 11:15	Compare as fractions: $\frac{7}{10} \approx 0.700$, $\frac{11}{15} \approx 0.733$, $\frac{5}{7} \approx 0.714$, $\frac{2}{3} \approx 0.667$
9	**(c)** 6:15:35	Make B equal: B $= \text{LCM}(5,3) = 15$; A $= 6$, C $= 35$ gives $6:15:35$
10	**(a)** 10	$25 \times 12 = 30 \times x$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$k = y/x = 63/9 = 7$ (1); $y = 7 \times 14 = 98$ (1).	1+1
12	$\frac{4}{5} = 0.8$ and $\frac{13}{17} = 0.76$ (1); so 4:5 is greater.	1+1
13	$xy = 12 \times 25 = 300$ (1); $y = 300 \div 10 = 30$ (1).	1+1
14	$k = y/x = 55/5 = 11$ (1); $y = 11 \times 8 = 88$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Length $= 30$ km (1); breadth $= 18$ km (1); perimeter $= 2(30+18) = 96$ km (1).	1+1+1
16	$1\% = 3.6^\circ$ (1); food 162°, rent 72° (1); education 18°, others 108° (1).	1+1+1
17	Speed × time is constant: $24 \times 35 = 840$ (1); new speed $= 840 \div 28$ (1) $= 30$ km/h (1).	1+1+1
	OR: Length $= 120$ km (1); breadth $= 70$ km (1); perimeter $= 2(120+70) = 380$ km (1).	1+1+1
18	$1\% = 3.6^\circ$ (1); food 72°, rent 144° (1); education 54°, others 90° (1).	1+1+1
	OR: Speed × time is constant: $50 \times 27 = 1350$ (1); new speed $= 1350 \div 18$ (1) $= 75$ km/h (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Make B equal: A:B:C $= 3:4:6$ (2); total parts $= 13$, one part $=$ ₹80 (1); A = ₹ 240, B = ₹ 320, C = ₹ 480 (2).	2+1+2
	OR: Angle per mark $= 360^\circ \div 45 = 8^\circ$ (1); angles: Walk 72°, Cycle 96°, Bus 120°, Car 48°, Train 24° (3); Walk percentage $= 20\%$ (1).	1+3+1
20	Make B equal: A:B:C $= 12:15:20$ (2); total parts $= 47$, one part $=$ ₹60 (1); A = ₹ 720, B = ₹ 900, C = ₹ 1200 (2).	2+1+2
	OR: Angle per mark $= 360^\circ \div 72 = 5^\circ$ (1); angles: Red 90°, Blue 120°, Green 60°, Pink 50°, White 40° (3); Red percentage $= 25\%$ (1).	1+3+1

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